UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Short notes and keywords

Short answers, a keyword index, rapid recall and a last-10-minutes page.

Class 009 · Introduction to maxima and minima for a function of two variables · compiled 15 October 2026

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1 Short notes

1 · Maxima, minima and saddle points

Local maxima are peaks, local minima are pits; at both the tangent plane is horizontal. A saddle point also has a horizontal tangent plane but rises one way and falls another. A function can have several of each.

2 · Definitions: relative maximum, minimum and extreme value

Relative maximum at (a, b): $f(a, b) > f(a + h, b + k)$ for small h, k; minimum: $<$. Equivalently $\Delta = f(a + h, b + k) - f(a, b)$ has one sign: negative ⇒ maximum, positive ⇒ minimum. Either is an extreme value.

3 · Conditions and the working rule

Solve $f_x = 0$, $f_y = 0$. At each point find $r = f_{xx}$, $s = f_{xy}$, $t = f_{yy}$. $rt - s^2 > 0$: $r < 0$ max, $r > 0$ min; $rt - s^2 < 0$: saddle; $rt - s^2 = 0$: doubtful (use Δ). Substitute the point in f for the value.

2 Keyword index

11 terms; the number in brackets is the topic.

r, s, t (3). f_{xx}, f_{xy}, f_{yy}.

Doubtful case (3). $rt - s^2 = 0$: test gives no information.

Extreme value (2). A maximum or a minimum value.

Hessian (3). $\begin{vmatrix} r & s \\ s & t \end{vmatrix} = rt - s^2$.

Horizontal tangent plane (1). Geometric meaning of a stationary point.

Local (relative) maximum (2). $f(a,b) > f$ at all nearby points.

Local (relative) minimum (2). $f(a,b) < f$ at all nearby points.

Neighbourhood (2). Points $(a+h, b+k)$ with h, k small.

Saddle point (1). Stationary, neither max nor min; $rt - s^2 < 0$.

Stationary / critical point (3). $f_x = f_y = 0$.

Working rule (3). Stationary points → r, s, t → sign of $rt - s^2$, then r.

3 Rapid recall

1	Relative maximum at (a, b)?	$f(a, b) > f(a + h, b + k)$ for all small h, k.
2	In terms of Δ?	$\Delta = f(a + h, b + k) - f(a, b)$ one sign: negative max, positive min.
3	Stationary points?	Solve $f_x = 0$, $f_y = 0$.
4	r, s, t?	f_{xx}, f_{xy}, f_{yy} at the point.
5	Maximum test?	$rt - s^2 > 0$ and $r < 0$.
6	Minimum test?	$rt - s^2 > 0$ and $r > 0$.
7	$rt - s^2 < 0$?	Saddle point.
8	$rt - s^2 = 0$?	Doubtful; needs further investigation.
9	What is the Hessian here?	$\begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix} = rt - s^2$.
10	Nature of $x^2 - y^2$ at the origin?	Saddle ($rt - s^2 = -4$).
11	$x^4 + y^4$ at the origin?	Test fails; minimum by Δ.
12	Min of $4x^2 + 2y^2 + 4xy - 10x - 2y - 3$?	$-\frac{23}{2}$ at $(2, -\frac{3}{2})$.

Mnemonic (study aid). *"Determinant first, then r."* Positive Hessian: r negative = hill (max), r positive = valley (min); negative Hessian = saddle; zero = doubt.

4 Last 10 minutes before the exam

Top facts

1. Max at (a,b): $f(a,b) > f(a+h,b+k)$ nearby; min: $<$. Extreme value = max or min.
2. Stationary points: $f_x = 0$, $f_y = 0$ (horizontal tangent plane).
3. $r = f_{xx}$, $s = f_{xy}$, $t = f_{yy}$; Hessian $rt - s^2$.
4. $rt - s^2 > 0$, $r < 0$: max. $rt - s^2 > 0$, $r > 0$: min.
5. $rt - s^2 < 0$: saddle point (not an extreme value). $rt - s^2 = 0$: doubtful.
6. Test every stationary point; substitute into f for the value.

Top formulas

E3.1 · Stationary points (necessary condition) ESA ★★★

$$f_x = \frac{\partial f}{\partial x} = 0, \qquad f_y = \frac{\partial f}{\partial y} = 0$$

GIVES The candidates $(a,b), (c,d), \ldots$ for extreme values.

APPLIES f differentiable; solve the two equations simultaneously.

NOTE Two-variable version of $\frac{dy}{dx} = 0$.

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E3.2 · Second derivatives used in the test ESA ★★★

$$r = \frac{\partial^2 f}{\partial x^2}, \qquad s = \frac{\partial^2 f}{\partial x\,\partial y}, \qquad t = \frac{\partial^2 f}{\partial y^2}$$

GIVES The three numbers at each stationary point.

APPLIES Evaluate at each stationary point separately.

NOTE $s = f_{xy} = f_{yx}$ (Clairaut).

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E3.3 · Hessian determinant ESA ★★★

$$rt - s^2 = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix} = f_{xx} f_{yy} - f_{xy}^2$$

GIVES Whether the stationary point is an extremum.

APPLIES At a stationary point.

NOTE Positive: extremum (decide by r). Negative: saddle. Zero: doubtful.

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E3.4 · The test (working rule, steps 3–6)

ESA ★★★

$$rt - s^2 > 0,\ r < 0 \ \Rightarrow\ \text{maximum}; \qquad rt - s^2 > 0,\ r > 0 \ \Rightarrow\ \text{minimum}$$
$$rt - s^2 < 0 \ \Rightarrow\ \text{saddle point}; \qquad rt - s^2 = 0 \ \Rightarrow\ \text{doubtful}$$

GIVES The nature of each stationary point.

APPLIES After E3.1–E3.3.

NOTE $r < 0$ plays the role of $\frac{d^2y}{dx^2} < 0$ in one variable. When $rt - s^2 > 0$, r and t have the same sign, so t can be used instead.

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